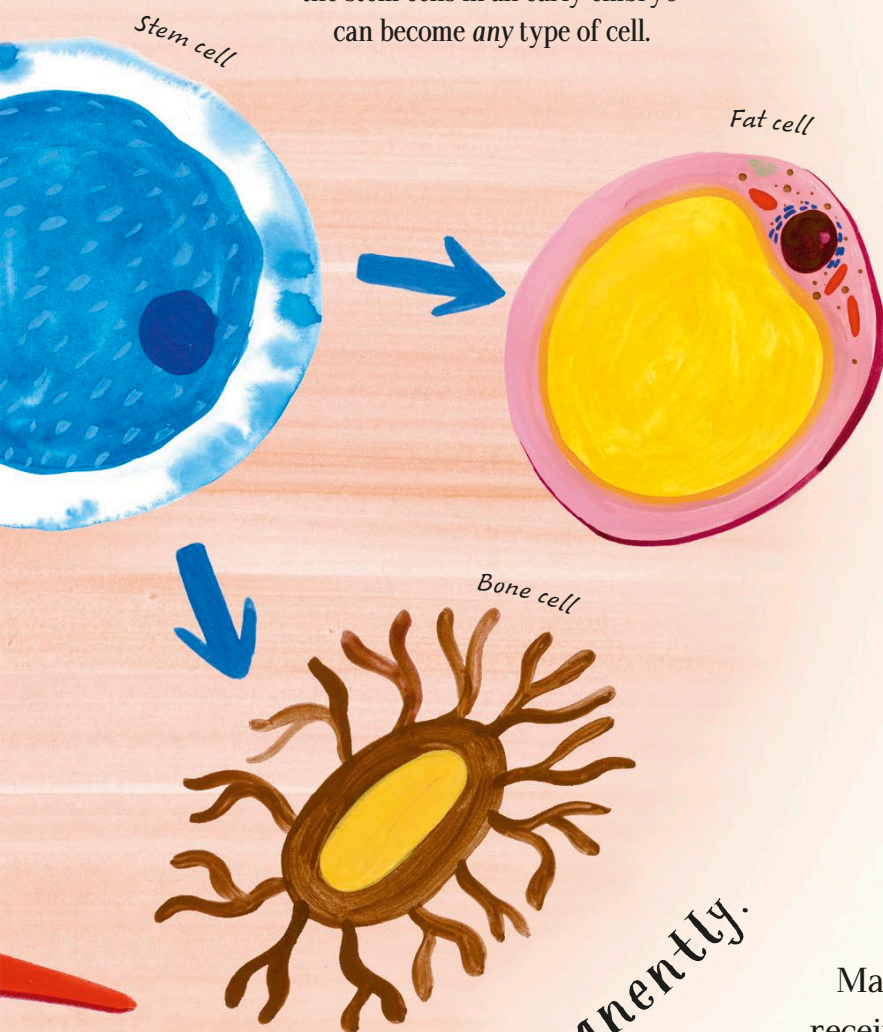
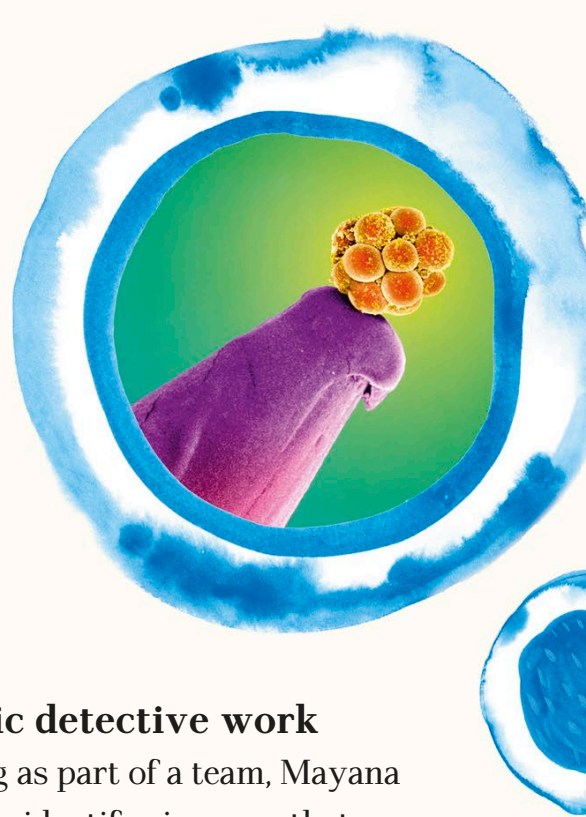


Stem cells

Most cells in our bodies are specialized—they have a certain shape and do a specific job. Stem cells are different. They have the potential to become different types of cell, so they can be used to replace cells that are diseased, damaged, or faulty. We all have some stem cells in our bodies, but only the stem cells in an early embryo can become any type of cell.



This is what a three-day-old human embryo looks like under a powerful microscope. It's balanced on the point of a pin! The cells inside are stem cells.



Genetic detective work

Working as part of a team, Mayana helped to identify six genes that can be passed from parents to children and may sometimes cause neuromuscular conditions. She worked closely with affected families and was shocked to find out how little support they had. Mayana founded a charity that helps children with neuromuscular conditions to access education, as well as wheelchairs and physical therapy.

Genetics research can help us to understand infectious diseases, too. Mayana's team has been investigating the Zika virus, which is carried and passed to humans by mosquitoes.

type of cell permanently.

Mayana also received funding to set up a center for genetics research at the University of São Paulo, in Brazil. There, her team looks for new ways to treat genetic diseases—for example, by using stem cell therapies. Mayana helps politicians and the public to understand the science behind stem cells, and how these new types of treatments could be used to save millions of lives in the future.

